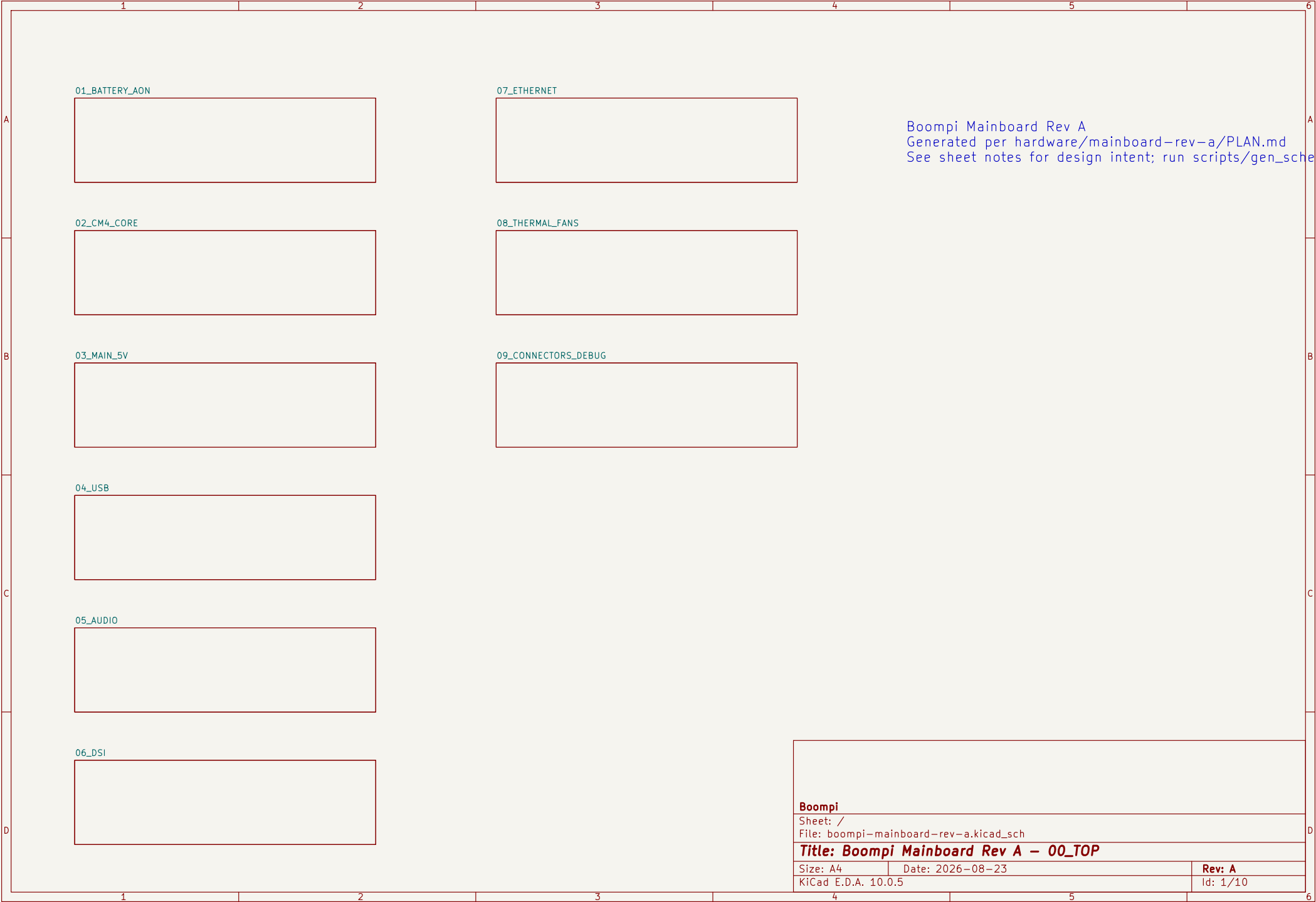
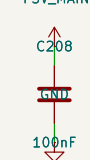
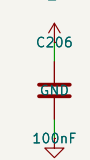
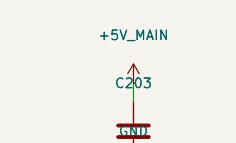
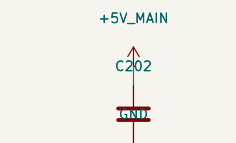
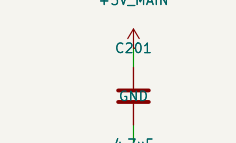
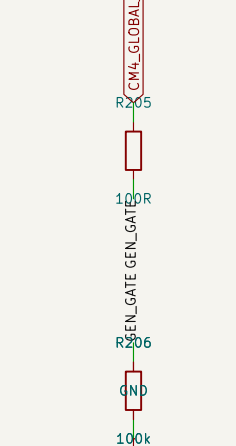
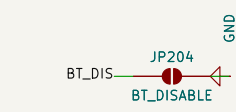
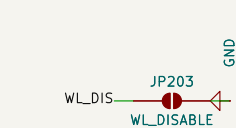
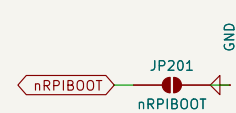
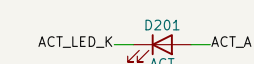
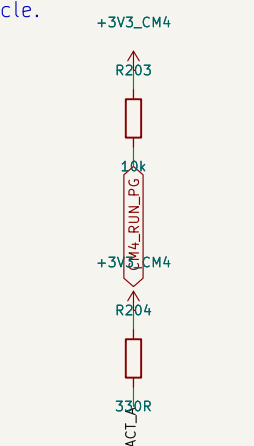
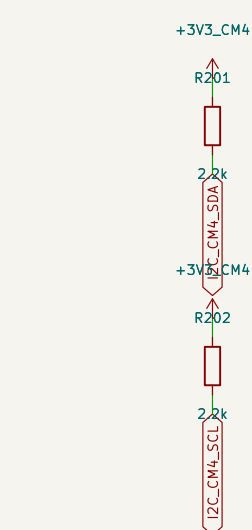
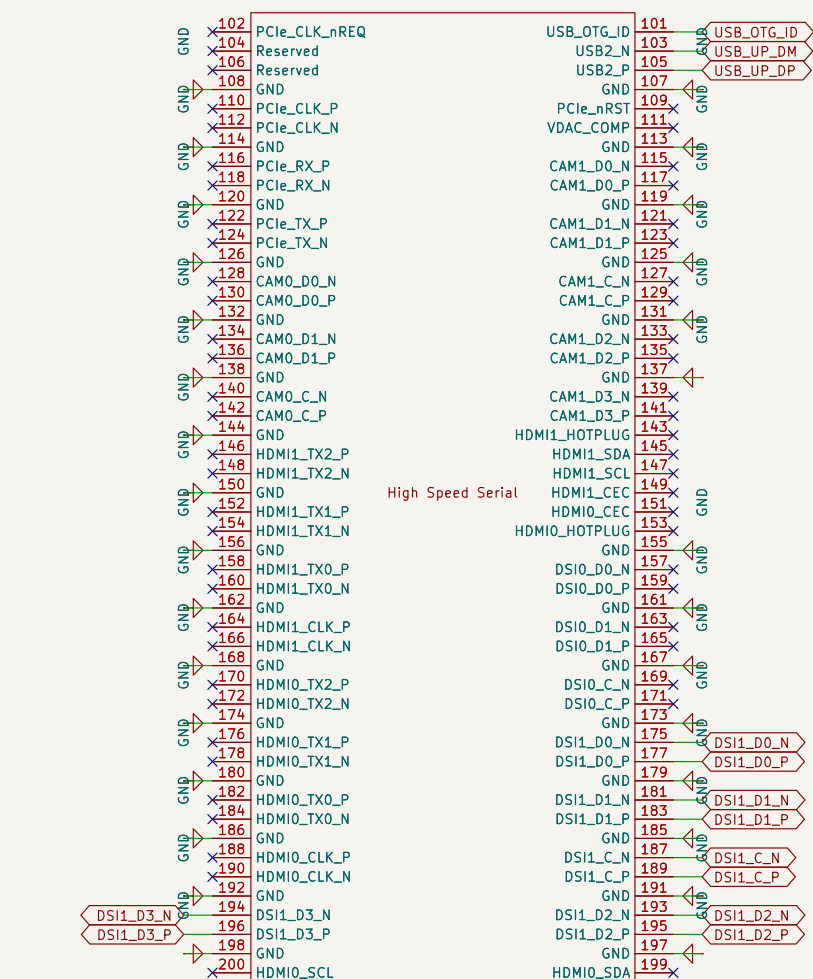
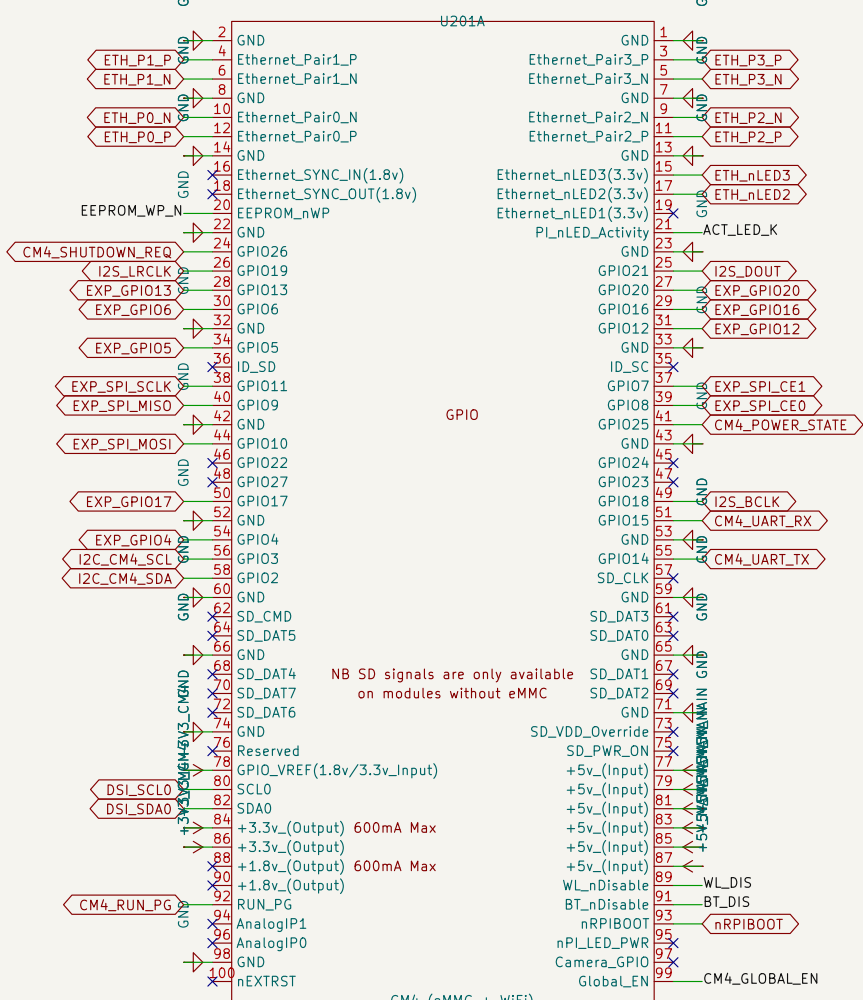
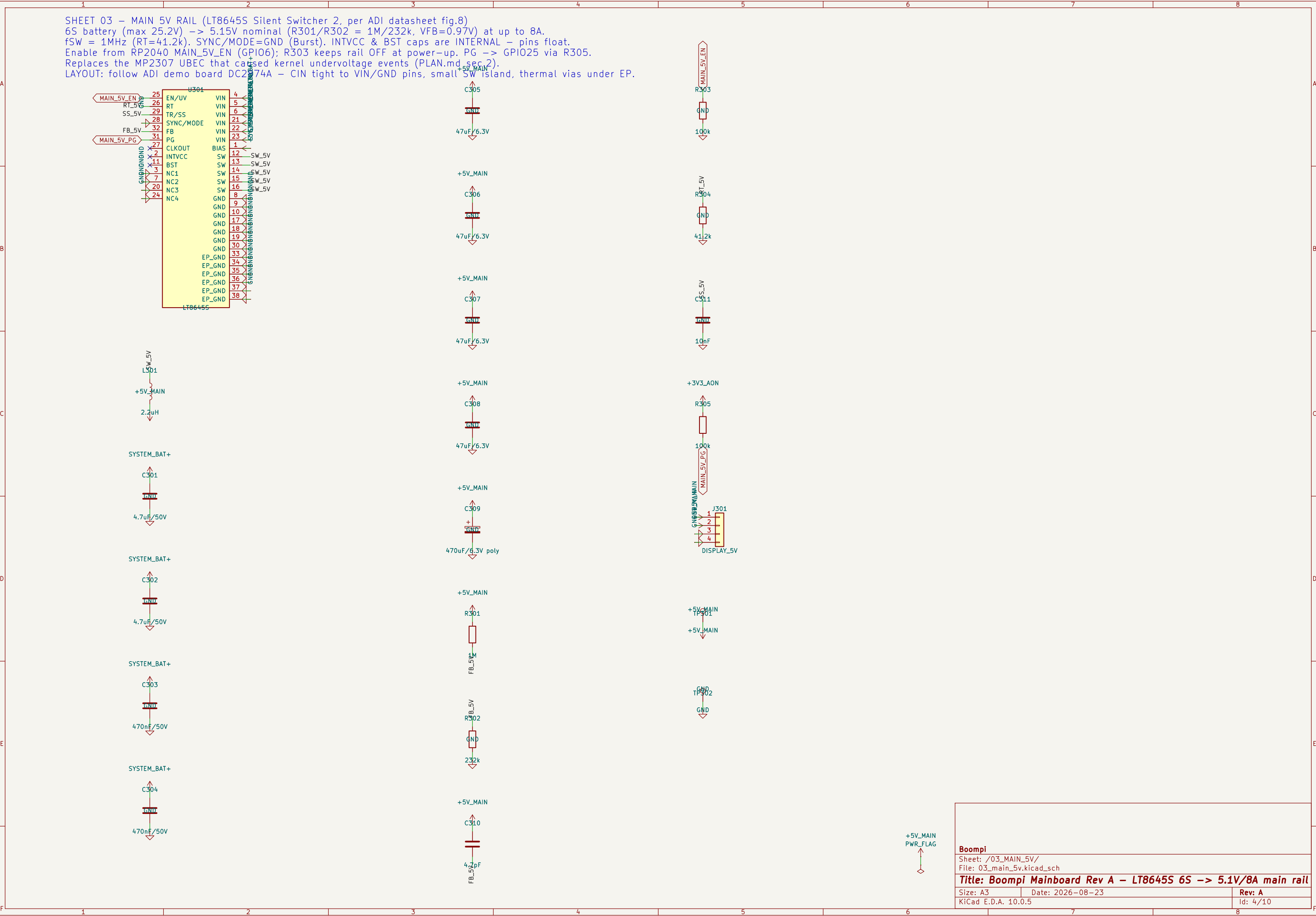




SHEET 02 – CM4 CORE (per Raspberry Pi CM4I0 reference design)
 eMMC CM4 variants: SD_* pins intentionally unconnected. HDMI / CSI / PCIe / composite unused in Rev A.
 DS11 (4-lane) is the Boomi display interface --> sheet 06. Onboard WiFi used; onboard BT NOT used for audio
 (UB500 on USB, sheet 04). JP201 nRPIBOOT: close to flash eMMC over the service micro-USB (also holds hub reset).
 Board_LEN has an internal module pull-up; Q201 lets the RP2040 force a hard power-cycle.
 RUN_PG pulled to 3V3_CM4 (R203) and sensed by RP2040 through R125 (sheet 01). +3V3_CM4



SHEET 03 – MAIN 5V RAIL (LT8645S Silent Switcher 2, per ADI datasheet fig.8)
6S battery (max 25.2V) -> 5.15V nominal (R301/R302 = 1M/232k, VFB=0.97V) at up to 8A.
fSW = 1MHz (RT=41.2k). SYNC/MODE=GND (Burst). INTVCC & BST caps are INTERNAL – pins float.
Enable from RP2040 MAIN_5V_EN (GPIO6); R303 keeps rail OFF at power-up. PG -> GPIO25 via R305.
Replaces the MP2307 UBEC that caused kernel undervoltage events (PLAN.md sec 2).
LAYOUT: follow ADI demo board DC274A – CIN tight to VIN/GND pins, small SW island, thermal vias under EP.

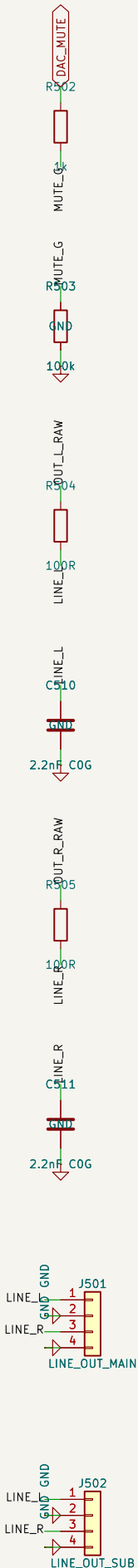
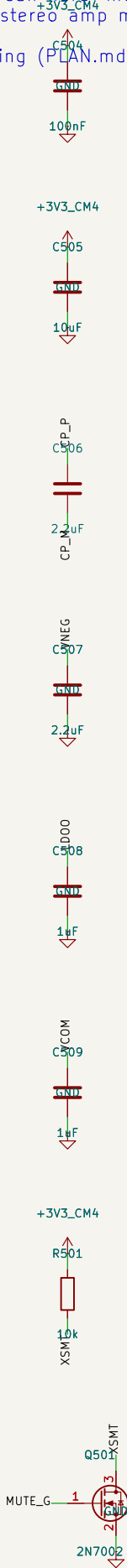
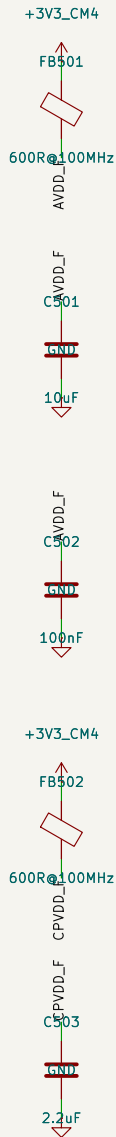
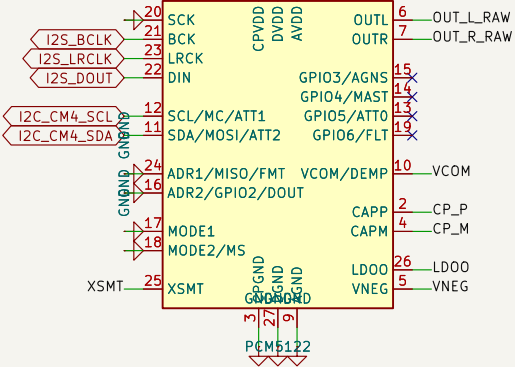


Boompi		
Sheet: /03_MAIN_5V/		
File: 03_main_5v.kicad_sch		
Title: Boompi Mainboard Rev A – LT8645S 6S -> 5.1V/8A main rail		
Size: A3	Date: 2026-08-23	Rev: A
KiCad E.D.A. 10.0.5		Id: 4/10

SHEET 04 - USB2514B 4-PORT HUB (PLAN.md sec.2)
CM4 USB2 -> hub upstream. Port1 -> internal USB-A (TP-Link UB500 Bluetooth). Port2 -> external USB-C data path to the PD board (J403; VBUS/CC handled on PD board). Port3 -> service header. Port4 -> spare (DNP). Straps: SMBus disabled (SDA/SCL 10k low) = default config; LOCAL_PWR_high = self-powered. OCS_N tied high (no per-port current sense IC; polyfuses instead). VBUS_DET from +5V_MAIN divider. rpiboot: close JP201 (sheet 02) -> CM4 rpiboot low AND hub RESET_N held low via D401; then the service micro-USB J402 talks through to the CM4 as a device. Keep J402 stub short in layout.

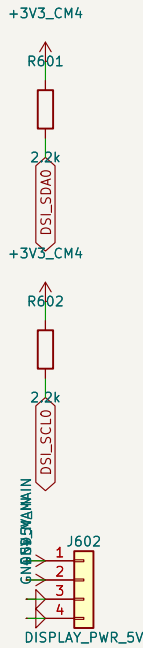
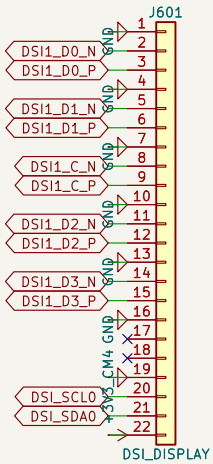


SHEET 05 – PCM5122 AUDIO DAC (PLAN.md sec.2)
I2S from CM4 (GPIO18/19/21), control via I2C_CM4 @ 0x4C (ADR1=ADR2=0). MODE1/MODE2=0 -> I2C mode.
SCK grounded: internal PLL generates the master clock from BCK (HiFiBerry-style, no external MCLK).
2.0 VRMS ground-centered line out (VNEG charge pump) -> no output caps, no ground-loop isolator.
XSMT: pulled up (unmuted) once 3V3_CM4 is up; RP2040 DAC_MUTE (GPIO29) can hard-mute via Q501
during power transitions for pop-free sequencing (PLAN.md sec.5). J501 -> stereo amp module,
J502 -> subwoofer amp module (module sums L+R).
LAYOUT: keep this section away from LT8645S / USB / Ethernet / fan switching (PLAN.md sec.9).

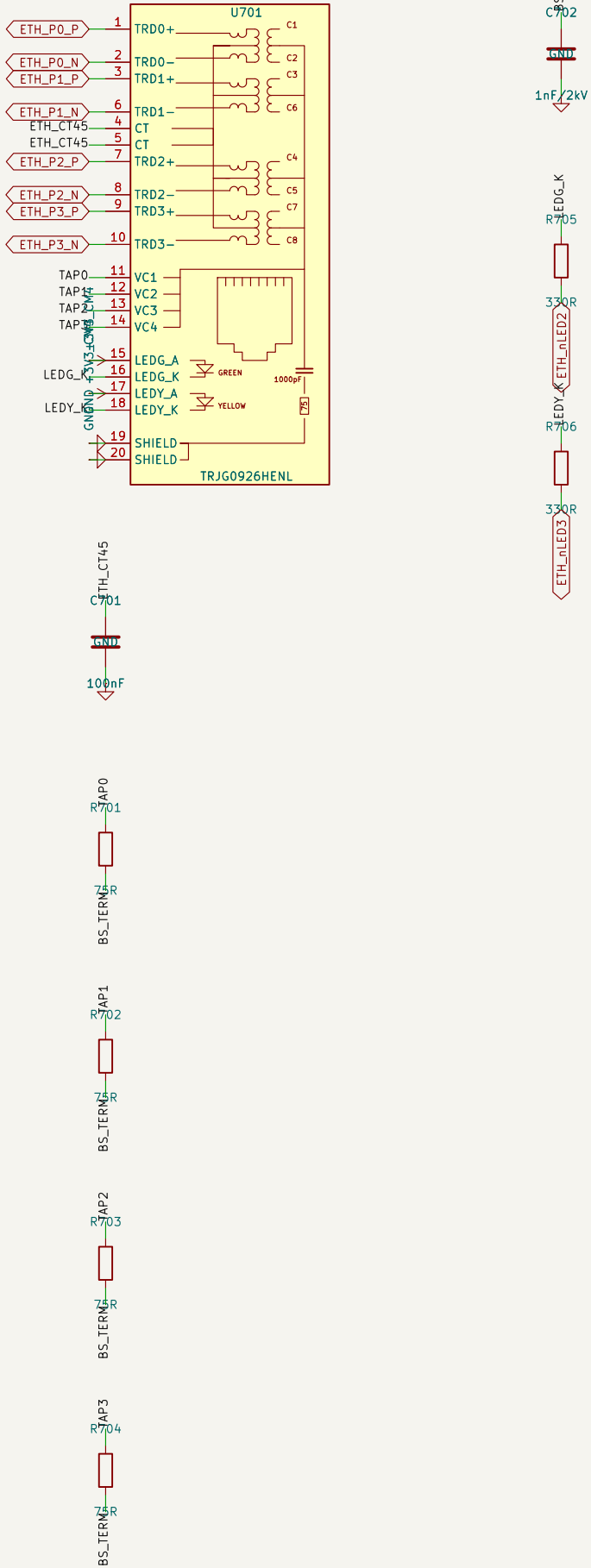


PWR_FLAG		
Boompi		
Sheet: /05_AUDIO/		
File: 05_audio.kicad_sch		
Title: Boompi Mainboard Rev A – PCM5122 I2S DAC, line outputs		
Size: A3	Date: 2026-08-23	Rev: A
KiCad E.D.A. 10.0.5		Id: 6/10

SHEET 06 – DSI DISPLAY (PLAN.md sec.2)
Raspberry Pi–standard 22–pin 0.5mm FFC on CM4 DSI1 – all 4 lanes wired so both 2–lane panels
(e.g. 4.3in OSOY00) and 4–lane panels (Touch Display 2) work; selection via device–tree overlay.
Pinout copied from CM4IO J16 (DISP1). Touch I2C = CM4 SDA0/SCL0 (pins 20/21), pulled to 3V3_CM4.
Display 5V comes from the separate keyed connector J602 (or J301 on sheet 03) – not through the FFC.
LAYOUT: 100R differential pairs, matched within pair, solid GND reference; keep < 100mm to connector.



SHEET 07 – GIGABIT ETHERNET (debug / rescue / development, PLAN.md sec.2)
CM4 has the PHY on-module (BCM54210) -> pairs go straight to an integrated-magnetics magjack.
Wiring copied from the CM4IO reference (same magjack family). Center taps: Bob Smith termination
75R x4 into 1nF/2kV. LEDs driven by CM4 Ethernet_nLED2 (link/green) / nLED3 (activity/yellow).
Some enclosures leave the RJ45 internally inaccessible – that is fine, nothing else depends on it.
LAYOUT: 100R differential pairs, pair-matched; void other signals under the jack.



Boompi

Sheet: /07_ETHERNET/
File: 07_ethernet.kicad_sch

Title: Boompi Mainboard Rev A – Gigabit Ethernet magjack

Size: A3

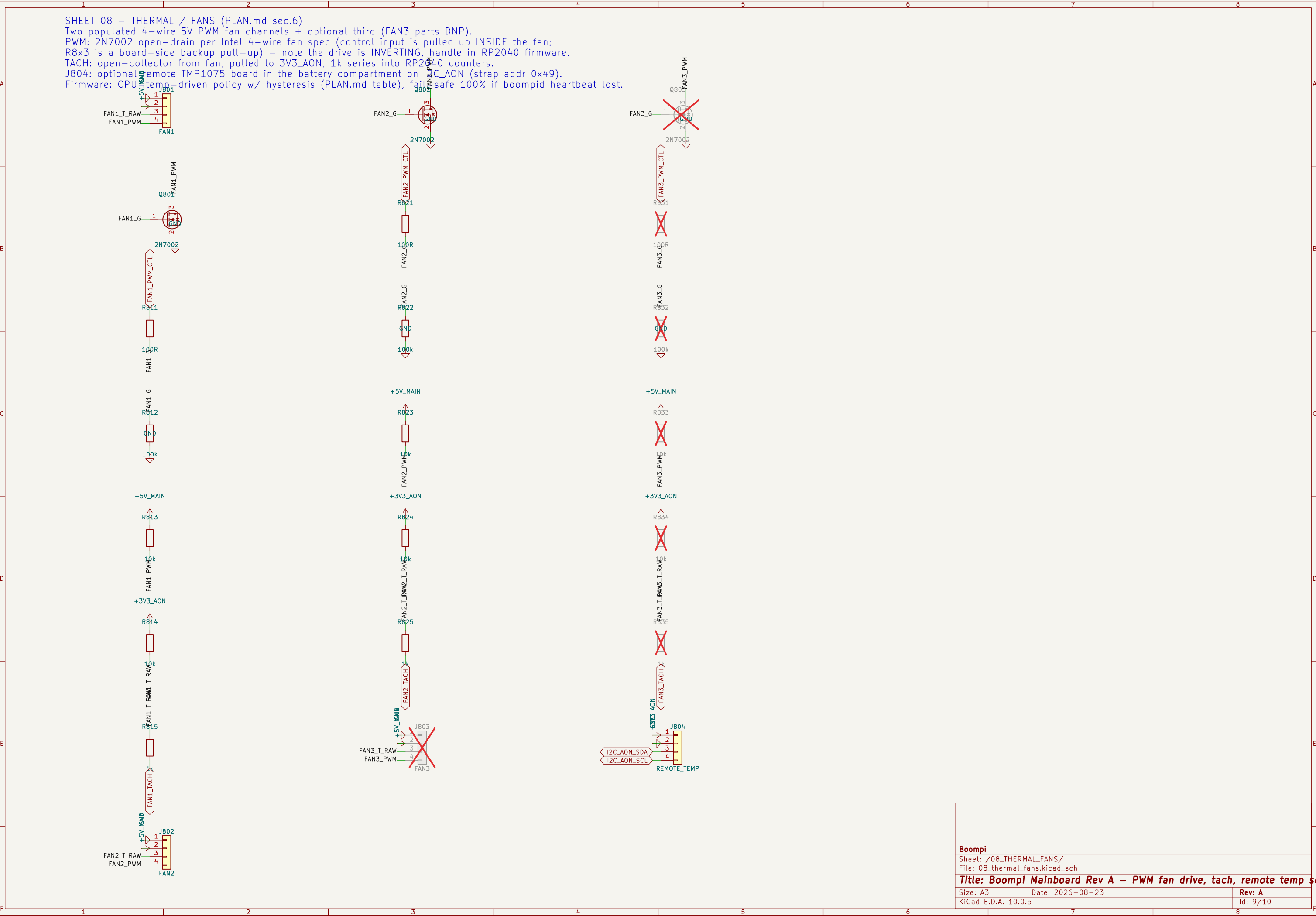
Date: 2026-08-23

Rev: A

KiCad E.D.A. 10.0.5

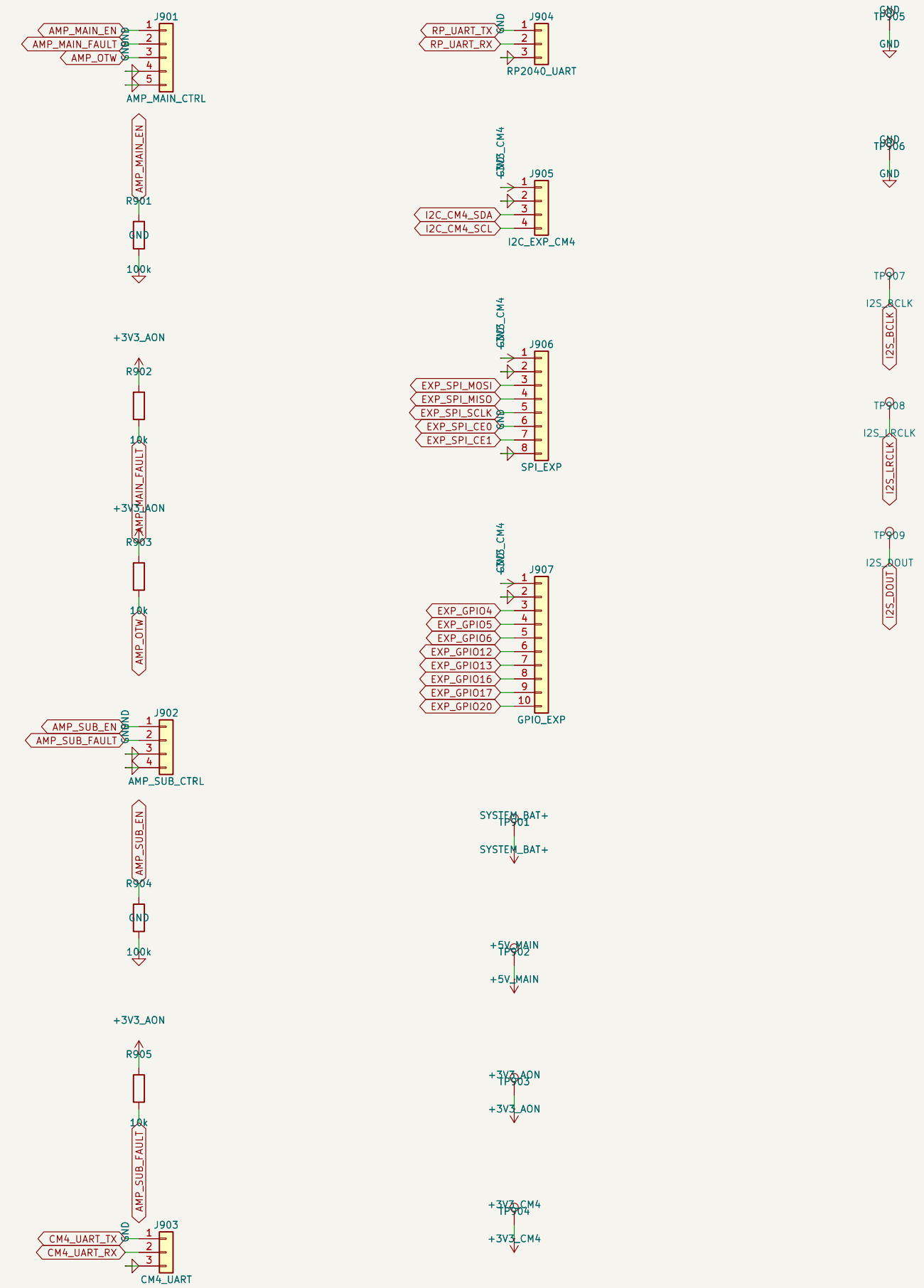
Id: 8/10

SHEET 08 – THERMAL / FANS (PLAN.md sec.6)
Two populated 4-wire 5V PWM fan channels + optional third (FAN3 parts DNP).
PWM: 2N7002 open-drain per Intel 4-wire fan spec (control input is pulled up INSIDE the fan;
R8x3 is a board-side backup pull-up) – note the drive is INVERTING, handle in RP2040 firmware.
TACH: open-collector from fan, pulled to 3V3_AON, 1k series into RP2040 counters.
J804: optional remote TMP1075 board in the battery compartment on I2C_AON (strap addr 0x49).
Firmware: CPU temp-driven policy w/ hysteresis (PLAN.md table), fail-safe 100% if boompid heartbeat lost.



Boomp		
Sheet: /08_THERMAL_FANS/		
File: 08_thermal_fans.kicad_sch		
Title: Boomp Mainboard Rev A – PWM fan drive, tach, remote temp sensor		
Size: A3	Date: 2026-08-23	Rev: A
KiCad E.D.A. 10.0.5		Id: 9/10

SHEET 09 – CONNECTORS / DEBUG (PLAN.md sec.7)
Amp module control per PLAN.md amp-module requirement (EN + FAULT + OTW):
J901 -> TPA3221 stereo module (EN default-off via R901; FAULT/OTW are open-drain, pulled to 3V3_AON).
J902 -> TPA3221 PBTl subwoofer module. Amp POWER comes from J103/J104 on sheet 01 (SYSTEM_BAT+).
Debug: CM4 console UART (GPIO14/15), RP2040 console UART (GPIO0/1), RP2040 SWD on sheet 01 (J106).
Expansion: CM4 SPI0 + 8 spare CM4 GPIOs + CM4 I2C. All 3V3 logic. Test points on every major rail.



Boompi		
Sheet: /09_CONNECTORS_DEBUG/ File: 09_connectors_debug.kicad_sch		
Title: Boompi Mainboard Rev A – Amp control, debug UART/SWD, expansion,		
Size: A3	Date: 2026-08-23	Rev: A
KiCad E.D.A. 10.0.5		Id: 10/10